

§ 191. Theorems from the theory of series.

[source p. 694]

For the further applications of the preceding results, some theorems from the theory of infinite series are required; these shall first be derived here.

1. Let $a_1, a_2, a_3, \dots, a_n, \dots$ be an unbounded system of real quantities (positive, negative, or also vanishing), of which we suppose that the sum

$$\sigma_n = a_1 + a_2 + \dots + a_n \quad (1)$$

remains, for every arbitrary n , below a finite bound C in absolute value. Then the series

$$S = \frac{a_1}{1^s} + \frac{a_2}{2^s} + \frac{a_3}{3^s} + \dots \quad (2)$$

is, for every positive s , not only convergent, but also a continuous function of s .

To prove this theorem, we decompose S in this way:

$$S = S_m + R_m,$$

where

$$S_m = \frac{a_1}{1^s} + \frac{a_2}{2^s} + \dots + \frac{a_{m-1}}{(m-1)^s},$$

$$R_m = \frac{a_m}{m^s} + \frac{a_{m+1}}{(m+1)^s} + \frac{a_{m+2}}{(m+2)^s} + \dots.$$

Now by (1) one has

$$a_m = \sigma_m - \sigma_{m-1}, \quad a_{m+1} = \sigma_{m+1} - \sigma_m, \quad a_{m+2} = \sigma_{m+2} - \sigma_{m+1}, \dots$$

and therefore

$$R_m + \frac{\sigma_{m-1}}{m^s} = \sigma_m \left(\frac{1}{m^s} - \frac{1}{(m+1)^s} \right) + \sigma_{m+1} \left(\frac{1}{(m+1)^s} - \frac{1}{(m+2)^s} \right) + \dots.$$

Since, by the assumption, $\sigma_{m-1}, \sigma_m, \sigma_{m+1}, \dots$ are enclosed between finite bounds $\pm C$, it follows that $R_m + \frac{\sigma_{m-1}}{m^s}$ is enclosed between the two bounds

$$\pm C \left(\frac{1}{m^s} - \frac{1}{(m+1)^s} + \frac{1}{(m+1)^s} - \frac{1}{(m+2)^s} + \dots \right) = \pm \frac{C}{m^s}$$

[source p. 695]

and hence, in absolute value,

$$R_m < \frac{2C}{m^s},$$

or, if $s > c$,

$$R_m < \frac{2C}{m^c}.$$

This upper bound for R_m , which is independent of s , can however, if c is positive, be depressed below any arbitrarily small value by taking m sufficiently large.

From this follows not only the convergence, but also the continuity of S . For if one takes m sufficiently large, then not only R_m , but also the fluctuation of R_m for variable s , becomes infinitely small, and S_m is, for a fixed m , a continuous function of s . Thus S too is continuous.

It is also to be observed that the assumption on σ_n by no means presupposes the convergence of the infinite series $\sum a_k$. It is not even necessary that the a_k be real; for if they are imaginary, one need only apply the theorem to the real and to the imaginary component. Finally it is also not necessary to assume $a_1, a_2, a_3, \dots$ to be constants. Everything remains valid when they are continuous functions of s .

2. Let $\mu_1, \mu_2, \dots, \mu_n, \dots$ denote an infinite set of positive numbers of such a kind that two finite numbers α, β can be given so that for every however large n

$$\alpha < \frac{n}{\mu_n} < \beta, \quad (3)$$

then the infinite series

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \frac{1}{\mu_3^s} + \dots$$

is convergent so long as the exponent s is greater than 1, and the product $(s-1)S$, as s approaches the value 1 sufficiently closely, likewise remains between the bounds α and β , or at least approaches one of these values without bound.

[source p. 696]

The proof is very simple to conduct by reduction to an integral. For plainly

$$\frac{1}{(n+1)^s} < \int_n^{n+1} \frac{dx}{x^s} < \frac{1}{n^s},$$

because the first or the second value, $(n+1)^{-s}$ or n^{-s} , is obtained when, under the integral sign, the too small or the too large value $(n+1)^{-s}$ or n^{-s} is put for x^{-s} .

Thus from (3), for every positive s , one obtains

$$\alpha^s \int_n^{n+1} \frac{dx}{x^s} < \frac{1}{\mu_n^s} < \beta^s \int_{n-1}^n \frac{dx}{x^s}.$$

If we put

$$R_m = \frac{1}{\mu_{m+1}^s} + \frac{1}{\mu_{m+2}^s} + \frac{1}{\mu_{m+3}^s} + \dots,$$

then this gives

$$\alpha^s \int_{m+1}^{\infty} \frac{dx}{x^s} < R_m < \beta^s \int_m^{\infty} \frac{dx}{x^s}$$

or

$$\frac{\alpha^s}{(s-1)(m+1)^{s-1}} < R_m < \frac{\beta^s}{(s-1)m^{s-1}}. \quad (4)$$

Thus R_m , however many terms may be summed, remains finite if $s > 1$, and for series with positive terms this is a sufficient condition for convergence.

If one further puts (4) in the form

$$\frac{\alpha^s}{(m+1)^{s-1}} < (s-1)R_m < \frac{\beta^s}{m^{s-1}},$$

one sees that the two bounds can be brought arbitrarily close to α, β if one takes $s-1$ small enough, and hence it follows that $(s-1)R_m$, for sufficiently small $s-1$, cannot lie outside

the bounds α, β unless, as $s - 1$ decreases, it approaches one of these values without bound. Since now R_n differs from S only by a component which is finite for every positive s , the theorem is thereby proved.

The theorem remains valid also when the inequality (3) does not hold for all n , but only holds as soon as n has exceeded an arbitrarily assigned bound m . On

[source p. 697]

the basis of this remark, if $n : \mu_n$ approaches a finite limiting value γ , one may take α and β arbitrarily close to this limiting value, and in this way obtains the following somewhat more precise form of the theorem:

3. If the positive numbers $\mu_1, \mu_2, \mu_3, \dots$ are so constituted that

$$\lim_{n=\infty} \frac{n}{\mu_n} = \gamma$$

is a finite limiting value, then the infinite series

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \frac{1}{\mu_3^s} + \dots$$

is convergent for every s greater than 1, and

$$\lim_{s=1} (s-1)S = \gamma.$$

Let now any system M of infinitely many positive numbers μ_n be given, and let us arrange them as

$$\mu_1 \leq \mu_2 \leq \mu_3 \leq \dots \quad (M), \quad (5)$$

so that in the series (5) a smaller term never follows a larger one; and let T denote the number of terms in M which are not greater than an arbitrarily given positive quantity t . Then the following theorem holds:

4. If one of the two limiting values

$$\lim_{n=\infty} \frac{n}{\mu_n}, \quad \lim_{t=\infty} \frac{T}{t} \quad (6)$$

is finite, then the other has the same finite value.

First let

$$\lim_{n=\infty} \frac{n}{\mu_n} = \gamma \quad (7)$$

be a finite limiting value. Then the μ_n must necessarily grow to infinity with infinitely increasing n , and for every positive t there is a value m such that

$$\mu_m \leq t < \mu_{m+1}; \quad (8)$$

m grows to infinity at the same time as t , and $T = m$ [by (5)]. Therefore by (8)

$$\frac{m}{\mu_m} \geq \frac{T}{t} > \frac{m}{\mu_{m+1}}. \quad (9)$$

[source p. 698]

If now $\gamma = 0$, then from this, letting t and m grow to infinity, it follows that $T : t$ also has the limiting value 0. But if γ is different from 0, then by (7)

$$\lim \frac{\mu_m}{\mu_{m+1}} = \lim \frac{m}{m+1} = 1,$$

and therefore the two limiting values in (9) approach the value γ , and hence it follows that

$$\lim_{t=\infty} \frac{T}{t} = \gamma. \quad (10)$$

Conversely, suppose secondly that the limit equation (10) is given. Then here too μ_n must grow to infinity with n , for otherwise, since the total number of all μ is infinite, T would already be infinite for a finite t , and γ could not be finite.

Now take a value μ_n which occurs in the series of mutually equal values

$$\mu_{m+1}, \mu_{m+2}, \dots, \mu_{m+l}$$

so that

$$m+1 \leq n \leq m+l.$$

Then T , when t passes through the value μ_n , will suddenly increase by l units, and the ratio $T : t$ increases by $l : \mu_n$. But since $T : t$ is to have a finite limiting value, one must have

$$\lim \frac{l}{\mu_n} = 0. \quad (11)$$

If now $t = \mu_n$, then $T = m + l$, and consequently

$$\frac{T}{t} = \frac{m+l}{\mu_n}, \quad \lim \frac{m+l}{\mu_n} = \gamma. \quad (12)$$

Further,

$$\frac{m}{\mu_n} < \frac{n}{\mu_n} \leq \frac{m+l}{\mu_n},$$

and consequently, by (11) and (12),

$$\lim \frac{n}{\mu_n} = \lim \frac{m}{\mu_n} = \gamma.$$

Thus equation (7) is derived from equation (10)¹.

Using theorem 4, we can now also give theorem 3 the form:

[source p. 699]

5. If T is the number of quantities μ_n that are not greater than t , then

$$\lim_{s=1} \sum_{n=1}^{\infty} \frac{s-1}{\mu_n^s} = \lim_{t=\infty} \frac{T}{t}, \quad (13)$$

provided that the limiting value of $T : t$ is finite.

¹This theorem is proved in essentially the same way in Dirichlet–Dedekind, Supplement II. Compare also Riemann’s mathematical works, second edition, No. XXX.

Assume that the quantities $\mu_1, \mu_2, \mu_3, \dots$ satisfy not only the conditions of theorem 3, but also the further conditions that, when

$$\mu_n = \frac{n + c_n}{\gamma} \quad (14)$$

is put, the c_n do not become infinite with infinitely increasing n (that is, for every however large n they are enclosed between finite bounds). Then theorem 3 can be replaced by the following sharper one:

6. If the numbers μ_n have the property that a finite γ can be determined so that

$$\gamma\mu_n - n = c_n \quad (15)$$

does not become infinite with infinitely increasing n , then the function of s defined for every $s > 1$ by

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \dots \quad (16)$$

has the property that the difference

$$S - \frac{\gamma}{s-1} = C \quad (17)$$

approaches a finite limit as $s-1$ decreases without limit.

To prove this, put, by (14) and (16),

$$\sum_{n=1}^{\infty} \frac{\gamma^s}{n^s} - S = \gamma^s \sum_{n=1}^{\infty} \frac{1}{n^s} \left[1 - \left(1 + \frac{c_n}{n} \right)^{-s} \right]. \quad (18)$$

If now ε is a positive proper fraction, then, putting

$$a_n = \frac{1}{n^\varepsilon} \left[1 - \left(1 + \frac{c_n}{n} \right)^{-s} \right],$$

the product $a_n n^{1+\varepsilon}$ does not become infinite for infinitely increasing n , and consequently, by a known elementary theorem of the theory of series, $a_1 + a_2 + \dots$ is an unconditionally convergent series.

[source p. 700]

Now put

$$s = s_1 + \varepsilon.$$

Then from (18) one obtains

$$\sum \frac{\gamma^s}{n^s} - S = \gamma^s \sum \frac{a_n}{n^{s_1}},$$

and this, by 1., is a finite and continuous function of s_1 for positive s_1 . But for $s_1 = 1 - \varepsilon$ one has $s = 1$, and consequently

$$\sum \frac{\gamma^s}{n^s} - S = D_s \quad (19)$$

is finite for $s = 1$, namely equal to

$$D_1 = \sum \frac{\gamma c_n}{n(n + c_n)}.$$

By application of simple theorems from the theory of the Γ -function, however, the sum $\sum \frac{1}{n^s}$ can be expressed by a definite integral. One has

$$\frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty e^{-nx} x^{s-1} dx,$$

and consequently

$$\sum \frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1} e^{-x} dx}{1 - e^{-x}}.$$

Further [by the theorem $(s-1)\Gamma(s-1) = \Gamma(s)$],

$$\frac{1}{s-1} = \frac{1}{\Gamma(s)} \int_0^\infty e^{-x} x^{s-2} dx,$$

and hence

$$\sum \frac{1}{n^s} - \frac{1}{s-1} = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} e^{-x} \left(\frac{1}{1 - e^{-x}} - \frac{1}{x} \right) dx.$$

Here the right-hand side is a continuous function for all positive values of s , and for $s = 1$ passes into Euler's constant

$$\int_0^\infty e^{-x} \left(\frac{1}{1 - e^{-x}} - \frac{1}{x} \right) dx = -\Gamma'(1) = 0.57721566 \dots$$

Now by (19)

$$D_s = \gamma^s \left(\sum \frac{1}{n^s} - \frac{1}{s-1} \right) - \left(S - \frac{\gamma^s}{s-1} \right),$$

[source p. 701]

and therefore

$$\lim_{s=1} \left(S - \frac{\gamma^s}{s-1} \right) = -\gamma\Gamma'(1) + D_1 = C_1, \quad (20)$$

which was to be proved, is finite².

§ 192. Application to the determination of the class number.

We now apply these theorems to the theory of the field Ω . By the numbers μ_n of theorem 5 we understand the norms of all ideals of the field Ω , and therefore by T the number of ideals in Ω whose norm is not greater than a given positive quantity t ; and we obtain

$$\lim_{s=1} \sum \frac{s-1}{N(\mathfrak{a})^s} = \lim_{t=\infty} \frac{T}{t}, \quad (1)$$

where the sum on the left-hand side extends over all ideals $\mathfrak{a}$ of the field. Now by § 153 the ideals split into a finite number of classes

$$A_1, A_2, \dots, A_h,$$

and the great goal is the determination of this number h , the class number.

The ideals $\mathfrak{a}_1$ of one of these classes A_1 are characterized by the fact that their products with one and the same integral functional φ_1 , belonging to the class A_1^{-1} , are principal ideals, that is,

$$\varphi_1 \mathfrak{a}_1 = \alpha$$

²The theorems on Γ -functions used here are found in the more detailed textbooks of integral calculus, for example Serret-Harnack, *Lehrbuch der Differential- und Integralrechnung*, vol. II, p. 170 f. (Leipzig 1885).

is an integral number of the field Ω .

If the norm of $\mathfrak{a}_1$ is not greater than t , then

$$N_a(\alpha) \leq N_a(\varphi_1)t = t_1.$$

If T_1 is the number of ideals of the class A_1 whose norms are not greater than t , then T_1 is at the same time the number of non-associated integral numbers divisible by φ_1 whose norms are not

[source p. 702]

greater than t_1 ; and by the theorem § 190, 1,

$$\lim_{t_1=\infty} \frac{T_1}{t_1} = \frac{g}{N_a(\varphi_1)}, \quad (2)$$

where g has the meaning given at the cited place, and is therefore a non-zero number determined by the nature of the field Ω .

If now $T_2, t_2, \dots, T_h, t_h$ have the corresponding meaning for the classes $A_2, \dots, A_h$, as T_1, t_1 have for A_1 , then

$$T = T_1 + T_2 + \dots + T_h, \\ \frac{T}{t} = \frac{T_1}{t_1} N_a(\varphi_1) + \frac{T_2}{t_2} N_a(\varphi_2) + \dots + \frac{T_h}{t_h} N_a(\varphi_h),$$

and from (1) and (2) there results the fundamental formula

$$\lim_{s=1} \sum \frac{s-1}{N(\mathfrak{a})^s} = gh. \quad (3)$$

The number g may be regarded as known by § 185, (11), even though its actual calculation still rests on the assumption that a fundamental system of units is known, and even though the determination of such a system in higher fields has always been regarded as one of the greatest difficulties. Then the calculation of the class number still depends on determining the limiting value on the left-hand side of (3), which of course succeeds only in special cases, but often permits important conclusions concerning the nature of the class number. We shall still develop a transformation of the sum

$$\sum \frac{1}{N(\mathfrak{a})^s} = \Phi(s) \quad (4)$$

into an infinite product, preparatory to the calculation.

Let $\mathfrak{p}$ be any prime ideal of f -th degree in the field Ω , so that

$$N(\mathfrak{p}) = p^f.$$

Then the infinite geometric series is

$$1 + \frac{1}{N(\mathfrak{p})^s} + \frac{1}{N(\mathfrak{p})^{2s}} + \dots = \frac{1}{1 - p^{-sf}};$$

if these series are multiplied together for all distinct prime ideals $\mathfrak{p}$, then, by known theorems from

[source p. 703]

the theory of infinite series, one obtains a sum of terms of the form

$$\left(\frac{1}{N(\mathfrak{p})^x} \frac{1}{N(\mathfrak{p}')^{x'}} \frac{1}{N(\mathfrak{p}'')^{x''}} \cdots \right)^s = \left(\frac{1}{N(\mathfrak{p}^x \mathfrak{p}'^{x'} \mathfrak{p}''^{x''} \cdots)} \right)^s,$$

which are all contained in the form $N(\mathfrak{a})^{-s}$, and every such term arises once and only once. Thus

$$\Phi(s) = \prod_{\mathfrak{p}} \frac{1}{1 - N(\mathfrak{p})^{-s}}. \quad (5)$$

If therefore $\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_e$ are the mutually distinct prime factors of p , and $f_1, f_2, \dots, f_e$ their degrees (§ 143), then

$$\Phi(s) = \prod_p \frac{1}{(1 - p^{-sf_1})(1 - p^{-sf_2}) \cdots (1 - p^{-sf_e})}, \quad (6)$$

where the infinite product $\prod$ is to be extended over all natural prime numbers p , and by (3), (4)

$$gh = \lim_{s=1} (s-1)\Phi(s). \quad (7)$$

From this one can draw a conclusion important for many applications:

If from the development

$$1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots = \frac{1}{1 - p^{-s}}$$

we form the product for all prime numbers p , then we get

$$\prod_p \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad (8)$$

where the sum on the right-hand side extends over all natural numbers n ; and thus this product has a finite value for all values of s greater than 1. The reciprocal value, namely the product

$$P = \prod_p (1 - p^{-s}), \quad (9)$$

whose factors are all smaller than 1, nevertheless has, so long as $s > 1$, a value different from zero. This property is preserved if in forming the product P we consider only a part of all prime numbers p , since the product is only increased by omitting arbitrary factors, without ever exceeding unity.

[source p. 704]

It follows that in the product (6), the partial products

$$\prod \frac{1}{1 - p^{-sf}},$$

which extend over all prime numbers p for which $f > 1$, remain finite even for $s = 1$. Since, on the other hand, g, h have definite non-zero values, the following important theorem follows from (7):

I. If $\mathfrak{p}$ runs through the totality of prime ideals of first degree of any algebraic field Ω , then the product

$$(s-1) \prod \frac{1}{1 - N(\mathfrak{p})^{-s}} \quad (10)$$

has, for $s = 1$, a finite limiting value different from zero.

This property is preserved also if, in forming the product, an arbitrary finite number of prime ideals of first degree is omitted.

The norms $N(\mathfrak{p})$ in formula (10) are equal to natural prime numbers p . But a prime number p occurs in it as often as it contains prime factors of first degree in Ω , hence at most n times. If Ω is a normal field, then every prime number p which is decomposable into prime factors of first degree actually occurs in it exactly n times, and for the product (10) we may also write

$$(s-1) \prod \frac{1}{(1 - p^{-s})^n},$$

where the product $\prod$ extends over all prime numbers p decomposable into prime factors of first degree (§ 160).

An immediate consequence of theorem I is:

In every algebraic field there are infinitely many prime ideals of first degree.

Twenty-fourth Section.

Class number of the cyclotomic fields.

§ 193. Class-number representation in the cyclotomic field Ω_m .

[source p. 705]

The general results concerning the class number h shall now be applied to the full cyclotomic field Ω_m , under the standing assumption that m is a power of a prime number q , and hence that

$$m = q^\alpha, \quad \varphi(m) = q^{\alpha-1}(q-1) = \mu \quad (1)$$

has been put.

In the twenty-first section (§ 169, 171) we saw that in q only one prime ideal of first degree occurs, and that a prime number p different from q and belonging to the exponent f decomposes into e distinct prime ideals of f -th degree. Here f denotes the smallest positive exponent for which

$$p^f \equiv 1 \pmod{m} \quad (2)$$

and e is determined by the equation

$$\mu = ef. \quad (3)$$

The function $\Phi(s)$ determined at the end of the preceding paragraph by formula (6) therefore here obtains the expression

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod \frac{1}{(1 - p^{-sf})^e}, \quad (4)$$

where the product sign $\prod$ extends over all prime numbers p different from q . Here s is a variable which is always greater than 1, and is finally to approach the bound 1.

[source p. 706]

In order now to transform this expression for the function Φ further and prepare it for calculation, one must use the theorems on Abelian groups and group characters which we learned in paragraphs 14 to 16 of this volume.

The totality of the numbers n not divisible by q , taken modulo m , forms an Abelian group $\mathfrak{R}$ of degree μ , whose characters $\chi(n)$ were determined in § 16. There was a small difference according as q is odd or $q = 2$.

1) If q is odd, then there is a primitive root c of m , and for every number n there is an index γ such that

$$n \equiv c^\gamma \pmod{m}. \quad (5)$$

If then Θ is a μ -th root of unity, then

$$\chi(n) = \Theta^\gamma, \quad (6)$$

and all μ characters are obtained by putting for Θ the different μ -th roots of unity.

If n belongs to the exponent f , then e is the greatest common divisor of γ and μ , and $\chi(n)$ is an f -th root of unity.

2) For $q = 2$, $m > 4$ (the case $m = 4$ is not considered here), every odd number n has two indices α, β , determined modulo $2, \frac{1}{2}\mu$, for which

$$n \equiv (-1)^\alpha 5^\beta \pmod{m} \quad (7)$$

holds. As characters one obtains, if $\varepsilon = \pm 1$ and Θ is put equal to a $\frac{1}{2}\mu$ -th root of unity,

$$\chi(n) = \varepsilon^\alpha \Theta^\beta; \quad (8)$$

f is the smallest positive solution of the two congruences

$$\alpha f \equiv 0 \pmod{2}, \quad \beta f \equiv 0 \pmod{\frac{1}{2}\mu},$$

and here also $\chi(n)$ is an f -th root of unity.

For both cases, however, the following remark holds:

3) If n belongs to the exponent f , then, when χ is allowed to run through the whole group of the μ characters, every f -th root of unity is obtained e times from $\chi(n)$.

For since the χ form an Abelian group, one first obtains each f -th root of unity which occurs at all equally often, namely as often as the value 1, since if $\chi_1(n) = \chi_2(n)$,

[source p. 707]

then $\chi_1 \chi_2^{-1}(n) = \chi_0(n) = 1$. On the other hand, however, every f -th root of unity is obtained; for if in case 1) Θ is taken to be a primitive μ -th root of unity, and in case 2) a primitive $\frac{1}{2}\mu$ -th root of unity, and in the latter case at the same time $\varepsilon = -1$, then $\chi(n)$ is a primitive f -th root of unity, and from its powers all f -th roots of unity can be derived.

If now x denotes a variable, then, when n belongs to the exponent f , the product extended over all characters χ is

$$\prod_{\chi} [1 - \chi(n)x] = (1 - x^f)^e, \quad (9)$$

and if one therefore puts $x = p^{-s}$, then from (4) one obtains

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod_{\chi} \prod_p \frac{1}{1 - \chi(p)p^{-s}}, \quad (10)$$

where the first product sign extends over all characters χ , and the second over all prime numbers p . Thus $\Phi(s)$ is a product of a finite number, μ , of factors, each of which is an infinite product.

We now expand each factor of one of these infinite products in increasing powers of p^{-s} . This gives

$$\frac{1}{1 - \chi(p)p^{-s}} = 1 + \chi(p)p^{-s} + \chi(p^2)p^{-2s} + \chi(p^3)p^{-3s} + \dots$$

The product of all factors so obtained, if χ is kept fixed while p runs through all primes different from q , is an aggregate of terms of the form

$$\chi(p^k)\chi(p'^{k'})\chi(p''^{k''})\dots p^{-sk}p'^{-sk'}p''^{-sk''}\dots = \chi(n)n^{-s},$$

and every term of this form occurs in the product once and only once [compare § 192, (8)]. Thus the transformation of $\Phi(s)$ into a finite product of infinite series is obtained:

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod_{\chi} \sum_n \frac{\chi(n)}{n^s}, \quad (11)$$

where the sum extends over all numbers n not divisible by q , and the product over all μ characters χ .

This expression is suited to determine the limiting value of $(s-1)\Phi(s)$ for $s=1$. Let us first consider the sum corresponding to the principal character $\chi=1$:

$$\sum_n \frac{1}{n^s}.$$

[source p. 708]

The numbers n are obtained by removing from the totality of all natural numbers k the multiples of q , that is, the totality of the numbers qk . Hence

$$\sum_n \frac{1}{n^s} = \sum_k \frac{1}{k^s} - \sum_k \frac{1}{q^s k^s} = (1 - q^{-s}) \sum_k \frac{1}{k^s}.$$

But if in theorem 3, § 191, we put for the μ_n the sequence of natural numbers k , then

$$\lim_{s \rightarrow 1} (s-1) \sum_k \frac{1}{k^s} = 1,$$

and therefore we obtain

$$\lim_{s \rightarrow 1} \frac{s-1}{1 - q^{-s}} \sum_n \frac{1}{n^s} = 1. \quad (12)$$

The other factors of the product $\Phi(s)$, however, are, when the infinite series are ordered according to increasing values of n , continuous functions of s by theorem 1, § 191.

For the sum ordered according to increasing values of n ,

$$\sum \chi(n), \quad (13)$$

is indeed not convergent, but nevertheless cannot go beyond a finite bound in absolute value. For as often as n runs through a complete residue system modulo m , a contribution is added to the sum (13) which, by § 11, 6, has the value 0. Hence, if χ is not the principal character,

$$\lim_{s \rightarrow 1} \sum_n \frac{\chi(n)}{n^s} = \sum_n \frac{\chi(n)}{n}, \quad (14)$$

and consequently, for the class number h in the cyclotomic field Ω_m , we now obtain the expression

$$gh = \prod_{\chi} \sum_n \frac{\chi(n)}{n}, \quad (15)$$

in which the product sign $\prod$ refers only to the characters χ different from the principal character.

§ 194. Determination of the sums X .

The infinite series

$$X = \sum_n \frac{\chi(n)}{n}, \quad (1)$$

[source p. 709]

on which, according to the preceding, the determination of the class number still depends, can be represented by finite expressions. Namely,

$$\frac{1}{n} = \int_0^1 x^{n-1} dx,$$

and therefore

$$X = \int_0^1 \sum_n \chi(n) x^{n-1} dx.$$

Now $\chi(n)$ remains unchanged when n is increased by a multiple of m . If, however, t denotes the least positive residue of n and one puts

$$n = t + ml,$$

then $\chi(n) = \chi(t)$, and it follows that

$$X = \int_0^1 \sum_t \chi(t) x^{t-1} \sum_{l=0}^{\infty} x^{ml} dx.$$

Put now

$$f(x) = \sum_t \chi(t) x^t. \quad (2)$$

Then $f(x)$ is an integral function of x of degree $m-1$, and at the same time, by the relation $\sum \chi(t) = 0$,

$$f(0) = 0, \quad f(1) = 0, \quad (3)$$

so that $f(x)$ is divisible by $x(1-x)$. For X , by the summation formula for the geometric series $\sum x^{ml}$, one obtains

$$X = \int_0^1 \frac{f(x) dx}{x(1-x^m)}. \quad (4)$$

To find such an integral, integral calculus prescribes decomposing the rational fraction under the integral sign into partial fractions. This partial-fraction decomposition gives, however, by (3), if

$$r = e^{\frac{2\pi i}{m}}$$

is put:

$$\frac{f(x)}{x(1-x^m)} = -\frac{1}{m} \sum_{s=1}^{m-1} \frac{f(r^s)}{x - r^s}$$

[Vol. I, § 29, (11)], and now we have further, by known

[source p. 710]

elementary theorems of integral calculus, if α denotes any angle between 0 and 2π ,

$$\int_0^1 \frac{dx}{x - e^{i\alpha}} = \frac{1}{2} \log \left(4 \sin^2 \frac{\alpha}{2} \right) + i \frac{\pi - \alpha}{2}.$$

It follows from this, by (4), that

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^s) \left[\frac{1}{2} \log \left(4 \left(\sin \frac{s\pi}{m} \right)^2 \right) - \frac{i\pi(m-2s)}{2m} \right]. \quad (5)$$

Here by (2)

$$f(r^s) = \sum_t \chi(t) r^{st}, \quad (6)$$

and hence [by (3)]

$$\sum_{s=1}^{m-1} f(r^s) = \sum_{s=0}^{m-1} f(r^s) = 0, \quad (7)$$

since

$$\sum_{s=0}^{m-1} r^{st} = 0.$$

Accordingly formula (5) simplifies somewhat further and gives

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^s) \left[\frac{1}{2} \log \left(\sin \frac{s\pi}{m} \right)^2 + \frac{i\pi s}{m} \right], \quad (8)$$

for which one may also, replacing s by $m-s$ and again using (7), put

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^{-s}) \left[\frac{1}{2} \log \left(\sin \frac{s\pi}{m} \right)^2 - \frac{i\pi s}{m} \right]. \quad (9)$$

From (6), however, follows

$$f(r^{-s}) = \sum_t \chi(t) r^{-st},$$

and the sum taken over t is not changed if t is replaced by $m-t$. Hence

$$f(r^{-s}) = \chi(-1) f(r^s).$$

The factor $\chi(-1)$ has the value $+1$ or -1 , since its square is $= +1$. We therefore now distinguish two kinds of characters $\chi_1(n), \chi_2(n)$, so that

$$\chi_1(-1) = 1, \quad \chi_2(-1) = -1 \quad (10)$$

[source p. 711]

and accordingly the functions f are also to be distinguished into f_1 and f_2 , so that

$$f_1(r^{-s}) = f_1(r^s), \quad f_2(r^{-s}) = -f_2(r^s). \quad (11)$$

Likewise we distinguish the expressions (8), (9) as X_1 and X_2 , and, adding the two expressions, we get by (11)

$$\begin{aligned} X_1 &= -\frac{1}{2m} \sum_{s=1}^{m-1} f_1(r^s) \log \left(\sin \frac{s\pi}{m} \right)^2, \\ X_2 &= -\frac{i\pi}{m^2} \sum_{s=1}^{m-1} s f_2(r^s). \end{aligned} \quad (12)$$